



A multiple level competitive swarm optimizer based on dual evaluation criteria and global optimization for large-scale optimization problem

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ABSTRACT

Large-scale optimization problems (LSOPs) in science and technology bring great challenges to the performance of algorithms. Although Competitive swarm optimizer (CSO) is an effective method, some shortcomings still exist when handling LSOPs, such as premature convergence. Therefore, a novel multiple level CSO with dual evaluation criteria and global optimization (DEGMCSO) is proposed to seek optimal solutions of LSOPs. In this paper, dual evaluation criteria are inserted into the multiple comparison process of the losers and winners to assist the algorithm retain more high quality particles with the potential. In addition to fitness values, adaptive selection weight fitness-distance is designed as the other criterion for selecting winners and losers according to different optimization problems. Meanwhile, a simple global optimal modification strategy is employed to get high quality global best solution. By CEC2010 and CEC2013 function suits, the results indicate DEGMCSO outperforms some popular algorithms. Finally, DEGMCSO is applied to feather selection problems of high dimension classification in the real world. The simulation results show that compared with the original CSO algorithm, DEGMCSO can find the solution of 16 functions on CEC2010 test function set which is obviously better than the CSO algorithm.

1. Introduction

Numerous complex optimization problems in science and technology have encompassed hundreds of variables [1], such as large-scale feature selection [2,3], deep learning network optimization [4], path planning [5], and some other practical problems [6,7]. These problems containing hundreds of variables can be taken regard as LSOPs [8]. In general, when the decision variable dimension is greater than or equal to 100, these problems are treated as high dimension problems, which can be expressed by following equation:

$$\begin{cases} \min/\max f(X), X = [x^1, x^2, \dots, x^D], D \geq 1000; \\ s.t. X \in \Omega \end{cases} \quad (1)$$

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where $\Omega = \{x \in R, l \leq x \leq u\}$ represents a continuous solution space between l and u . X represents decision variables. $f(X)$ represents objective function. The dimension increase of the decision variable will result in “dimensional disaster” problem.

Evolutionary computation algorithms (ECs) have gained a considerable success in dealing with a lot of complex small-scale optimization problems with fewer than a hundred variables. Compared with the traditional optimizers, ECs have many unique advantages, such as global optimization, parallelism, robustness [9], etc., and been primarily designed to handle many complicated engineering problems [10]. Conventional evolutionary computation algorithms, including ant colony algorithm [11], particle swarm optimization [12], etc., have certain advantages to handle low-dimensional global optimization problems [13]. However, in the face of LSOPs, there are some problems such as local extremum, general optimization ability and low convergence accuracy. The search effect and efficiency of these algorithms will decline sharply when solving LSOPs.

In recent years, the approaches to solving LSOPs problem mainly are divided into two categories. One type is EC algorithms by exploiting cooperative co-evolution (CC) framework, where LSOPs is decomposed into small sub-problems, and then solve them independently. A genetic algorithm based on co-evolution (CCGA) is proposed in 1994 [14], which is the first cooperative co-evolutionary algorithm (CCEC) based on CC framework. Since the optimization performance of CCGA may even decline for completely inseparable problems, based on the different grouping strategies, a number of corresponding improvement methods based on CC framework are developed, such as CCPSO [15], DECC [16] and MLCC [17]. CCECs display some advantages in dealing with LSOPs, but there are still some shortcomings. It is difficult to avoid consuming too much FEs, so that more FEs can be used to obtain better performance. Moreover, co-evolutionary algorithms rely heavily on decomposition strategies, and the effectiveness of decomposition strategies will impact the effect of algorithms, further limit the application of algorithms.

To address the above limitations, numerous non-cooperative co-evolution approaches have been made to handle LSOPs, unlike CCECs, which take all variables in the LSGO problem as a whole, rather than decomposition optimization. It is committed to developing new learning and updating strategies to maintain good diversity and thus assist traditional evolutionary algorithms cope with LSOPs. Some algorithms enhance population diversity by employing hybrid algorithms (e.g. MOGA [18]) or dynamic hierarchical learning to update the offspring (e.g. DLLSO [19]). While others integrate multiple different local search strategies in the population (e.g. jDElsco [20], SHADE-ILS [21], MA-SW-Chains [22]) to improve search accuracy. In addition, the dynamic multi-population population structure and university information interaction strategy [23] are designed to handle LSGOs. Although the algorithms improve the accuracy of indivisible LSGO problems to some extent, the huge search space still poses challenges to the accuracy of the solution.

Because the decision space expands exponentially with the increase of variable dimension, the efficiency of existing algorithms in solving LSOPs decreases. The local optimal trap is a difficult problem to solve for a large and complex search space, which may be handled by using efficient selection or mutually supportive powerful search operations [24]. It is difficult to succeed in selecting the elite particle method by relying only on the fitness. In order to avoid premature convergence, it should be determined by solution candidate with greatest contribution, but the researchers have paid less attention to choice. The selection method can help the algorithm find valuable solution in the search space [25]. Selecting excellent particles to guide population evolution through fitness values alone may lead to local optimality. Traditional algorithms are unable to select better individuals in LSOPs, and are unable to explore the huge decision space through basic operations, which often leads to algorithms finding sub-optimal solutions in limited local space. In the case of the selection, the selected individual usually does not contribute significantly to the solution of the problem. As a result, this situation has forced the development of special individuals that can guide the evolution of entire populations and explore vast decision spaces.

Therefore, this paper presents a multiple level CSO with dual evaluation criteria and global optimization. In the improved CSO based on multiple comparisons, dual evaluation criteria are introduced to select excellent particles. The first criterion based on fitness value is used to distinguish the losers and winners in the first comparison. In the subsequent comparison, the other criterion based on the score of the distance and fitness value, is adopted to choose the winners. Compared with a single evaluation criterion based only on fitness values, dual-criteria strategy can make full use of the two criteria to select valuable particles to guide the population evolution so that the algorithm can obtain better searching ability. In the second criterion, by introducing distance into search and dynamically selecting proportion of distance according to the evolutionary problem, this way can make sure the algorithm finds more particles with good diversity.

The main contributions of this study are listed as follows:

(1) A dual-criteria evaluation mechanism is designed to optimize the particle selection quality of multiple competition CSO algorithm. This evaluation mechanism can complement the shortcomings of existing selection mechanisms and guide the population to retain more valuable solutions.

(2) A novel adaptive weighted fitness-distance is proposed, which will provide a proper comparison criterion to select the winners and losers. Adaptive weights selection method are adopted to choose the appropriate weight for different optimization problems.

The remainder of the paper is organized as follows. In Section 2 briefly reviews the related work. Section 3 describes the proposed algorithm in detail. The experimental results are used to test the performance of DEGMCSO in Section 4. Finally, Section 5 drawn the paper and the future work.

2. Related work

2.1. Traditional PSO algorithm

PSO is an efficient global optimization algorithm simulating the behavior of birds. Each particle with corresponding fitness value is given two attributes of location and velocity. Each particle is updated as follows:

$$V_i(t+1) = \omega V_i(t) + c_1 r_1(t)(pbest_i(t) - X_i(t)) + c_2 r_2(t)(gbest(t) - X_i(t)) \quad (2)$$

$$X_i(t+1) = X_i(t) + V_i(t+1) \quad (3)$$

where $pbest_i(t)$ represents historical best position of individual i ; $gbest(t)$ represents the position of global best individual in population so far; t is the generation number, ω is the inertia weight, c_1 and c_2 are the acceleration coefficient, r_1 and r_2 are the random number in $[0,1]$.

2.2. PSO variants for LSOPs

In recent years, numerous PSO variants algorithms for LSOPs have been presented. Different from the CCGA algorithm, the CCPSO algorithm decomposed multiple dimensions into the same subgroup. Its optimization performance on most test functions is better than that of CCGA, but it can't solve the optimization problem with correlation between variables. Aim to increase population diversity and avoid local optimality, Li et al. [15] presented CCPSO2 algorithm, which adopted local neighborhood topology update rules based on Gauss and Cauchy to enhance search capability. Conventional decomposition methods are composed by dynamic decomposition method and static decomposition. Despite the considerable success of CCPSOs, they are highly sensitive to decomposition strategies and a substantial quantity of FEs are indispensable, particularly in scenarios where the scale of the problem is vast.

Considering the above limitations of CC, a dynamic multi-group algorithm with local search technology is presented. A sort-based biased learning PSO algorithm (RBLSO) is designed [26]. To enhance population diversity, two PSO variations: Competitive learning particle swarm algorithm (CSO) are proposed. CSO utilizes a competitive mechanism to select the winners and losers from the current individuals. Mohapatra et al. [27] adopted an improved CSO method (MCSO) in which two groups of particles were selected from three groups of particles to generate new positions through three groups of competition, which could accelerate the convergence rate. Lan et al. [28] designed a new competitive group optimizer based on two-stage learning (TPLSO). A hierarchical learning group optimization method (DLLSO) was designed based on dynamic layering based on fitness values. Wang et al. [29] proposed PSO algorithm controlled by providing reinforcement learning (RLLPSO) strategy for level number control. Although numerous excellent algorithms have been emerged, the diversity and convergence speed still need to be further coordinated by some new strategies.

3. Proposed method

3.1. Motivation

CSO is to mitigate the impact of global optimal solutions, thereby avoiding premature convergence. Compared with PSO, CSO features two important improvements. First, in PSO, the particle is guided by global best solution, but in CSO, instead of $gbest$, the search is propelled by pairwise competition among the particles, so any particle can become the leader. Second, there is no history in CSO, and particles that fail to compete only learn from those winners in the current population. However, CSO algorithm also has its deficiency. In CSO, because the group judges the winners or losers of the swarm only through a random comparison of fitness values, and then losers learn from winners. This selection method has some difficulty to guarantee the accuracy of losers and winners, so it will affect the quality of new individuals. To overcome the shortcoming, we provide a novel comparison criterion by incorporating the distance into the comparison process, the weight fitness-distance is used to balance the fitness criterion and distance criterion. Meanwhile, in each iteration, the optimal solution optimization strategy is used to get accurate solution. DEGMCSO is described as follows.

3.2. The proposed DEGMCSO

3.2.1. Multiple level CSO with adaptive weighted fitness-distance

In CSO, the single criterion based on fitness value does not consider the distance of the particles. Therefore, adaptive weight fitness-distance criterion is put into the comparison. This selection method not only considers the fitness values, but also considers distance of each particle from global best individual, which creates diverse individual. The evaluation function can be expressed as follows:

(1) Fitness-Based score

$$PF_i = 1 - \frac{f_i - f_{\min} + \varphi}{f_{\max} - f_{\min} + \varphi} \quad (4)$$

where f_i is the fitness of i -th particle. $f_{\min}$ and $f_{\max}$ are the maximum and minimum fitness values of all particles, respectively. Considering that $f_{\min}$ may be equal to $f_{\max}$, so φ is set as a small positive number and $\varphi = 1E - 5$.

(2) Distance-Based score

$$d_i = \sqrt{(x_i^1 - x_{best}^1)^2 + (x_i^2 - x_{best}^2)^2 + \dots + (x_i^D - x_{best}^D)^2} \quad (5)$$

$$D_i = \frac{d_i - d_{\min} + \varphi}{d_{\max} - d_{\min} + \varphi} \quad (6)$$

where d_i denotes the distance between particle i and global best individual, its calculation formula is given by formula (5), $d_{\min}$ and $d_{\max}$ are maximum and minimum values of d_i , respectively. D_i is the score based on distance. Considering that $d_{\min}$ may be equal to $d_{\max}$, $d_{\max}$ is set as a small positive number and equal to $1E - 5$.

(3) Score based on weighted fitness-distance

$$\omega = \text{Gaussian}(\mu, \sigma^2) \quad (7)$$

$$E_i = \omega \cdot PF_i + (1 - \omega) \cdot D_i \quad (8)$$

where ω represents the weight of the fitness value, which is determined by a Gaussian distribution. Its mean value μ is chosen from a candidate pool $S = [0.1, 0.2, 0.5, 0.8]$, and the variance $\sigma^2 = 1/12$. $(1 - \omega)$ is the weight of the distance score. PF is the fitness-based score, which is calculated by Eq.(4).

In the multiple level competitive swarm optimizer, all the particles are randomly decomposed into two parts. Then select two individuals from each group randomly and compare their fitness. After all the particles have been compared, the first comparison of competition is completed. We generate the winners and losers of the first comparison. In order to obtain high-quality excellent particles with good diversity, the particles that win in the first comparison will continue to participate the second competition according to score E_i based on weighted fitness-distance. Through the second comparison, the particle that wins in the second comparison have better fitness and diversity, which will be regarded as the elite particles. Then we pick up the lost particles in first comparison and make a third comparison according to the total score E_i based on weighted fitness-distance. The losers in third comparison are obtained, then we go on to compare these losers and find the real inferior particles as the losers by M comparisons. The losers in the 2nd competition or winners in the 3rd competition will be kept and enter into the next cycle. Fig.1 displays the comparison process.

These losers will be updated by two-phase strategy, different positional upgrade methods are built for each stage, which can obtain better global exploration ability. The number of fitness evaluations is expressed as FE. In the first stage, when current fitness evaluations $FEs < FE_{s1}$ ($FE_{s1} = 2/3 \text{ MaxFEs}$, MaxFEs is maximum fitness evaluations). The losers are updated normally by Eq.(10), and in the second stage, when current fitness evaluations $FEs > FE_{s1}$, the vibration is brought into the update process of the losers' position by Eq. (11). In DEGMCSO, only the losers after M comparisons need to update their positions, other particles will be retained for the next generation.

$$V_{C_L^M}(t+1) = r_1(t)V_{C_L^M}(t) + r_2(t)(X_{C_{WR_1}^2}(t) - X_{C_L^M}(t)) + \varphi_1(X_{C_{WR_2}^2}(t) - X_{C_L^M}(t)) \quad (9)$$

$$X_{C_L^M}(t+1) = X_{C_L^M}(t) + V_{C_L^M}(t+1) \quad (10)$$

$$X_{C_L^M}(t+1) = X_{C_L^M}(t) + 0.8V_{C_L^M}(t+1) + 0.3r_3(t)V_{C_L^M}(t+1) \quad (11)$$

where $V_{C_L^M}(t)$ is velocity vectors of the losers after M competitions, $X_{C_{WR_1}^2}(t)$ denotes $\lceil \frac{NP}{2^{M-1}} \rceil$ position vectors randomly selected from the

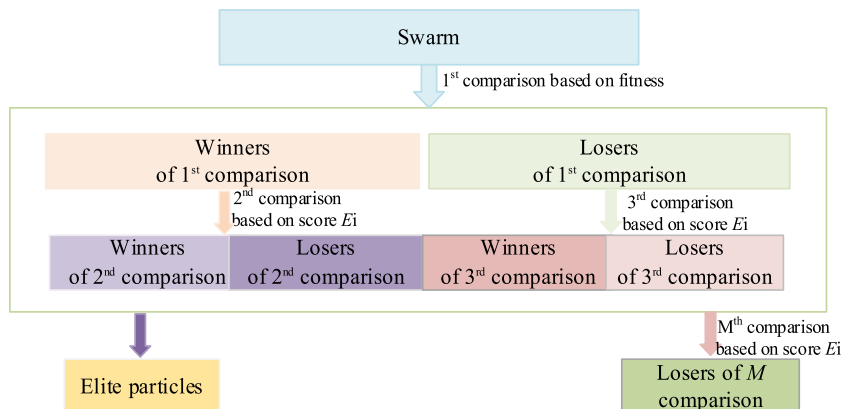


Fig. 1. The M comparisons based on dual evaluation criteria.

$\lceil \frac{NP}{2} \rceil$ winners in two competitions, $X_{L_M}(t)$ represents losers' position vectors after M competitions, $X_{C_W^2 R_2}(t)$ denotes the $\lceil \frac{NP}{2^{M-1}} \rceil$ position vectors randomly selected from the $\lceil \frac{NP}{2} \rceil$ winners in the second competition again; $r_1(t), r_2(t)$ and $r_3(t) \in [0, 1]^D$; φ_1 is the control factor, which controls the effect of $X_{C_W^2 R_2}(t)$.

The weight is to regulate diversity and convergence of DEGMCSO. It is determined by the performance of each weight in search process. Adaptive selection weight fitness-distance selects the appropriate weight according to the change rate of fitness value in the process of evolution. The adaptive weight selection method is given as follows:

(1) Create a set $S = [0.1, 0.2, 0.5, 0.8]$, before the number of iterations does not reach G_2 , each iteration randomly chooses a weight value from the set S .

(2) Then, we define a list of performance records $R = [R_1, R_2, R_3, R_4]$. Each ω in S is connected to R_i . When iterations number is G_2 , R_i is mathematically expressed as:

$$R_i = \sum r_i / N_i \quad (12)$$

where N_i is time number, ω_i is selected in the specified iteration number G_2 , here G_2 is set to 500, r_i is the change rate of fitness value when $\omega = \omega_i$, its value can be represented as:

$$r_i = \frac{|F_g(t) - F_g(t-1)|}{|F_g(t-1)|} \quad (13)$$

where $F_g(t)$ and $F_g(t-1)$ are the fitness value of the global best particle P_g in the t -th and $(t-1)$ -th iteration, respectively.

(3) According to Eq. (12), the R_i corresponding to all available ω_i in the set S is calculated, and ω_i corresponding to the maximum R_i is used as weight value for subsequent iteration operation.

3.2.2. Global optimization

For LSOPs, due to the large number of variables, some low-quality dimensions will limit the quality of the best individual. Therefore, it is very important to optimize the bottleneck dimension of the best individual. Global optimization strategy is designed to modify the bottleneck dimension can facilitate the evolution of the algorithm. Monte Carlo method randomly selects j -th dimension from the global optimal solution, and then the particle is randomly chosen from the winners of two competitions, namely elite particles. The j -th dimensions of these elite particles are used to replace j -th dimension of the global optimal solution, thereby producing

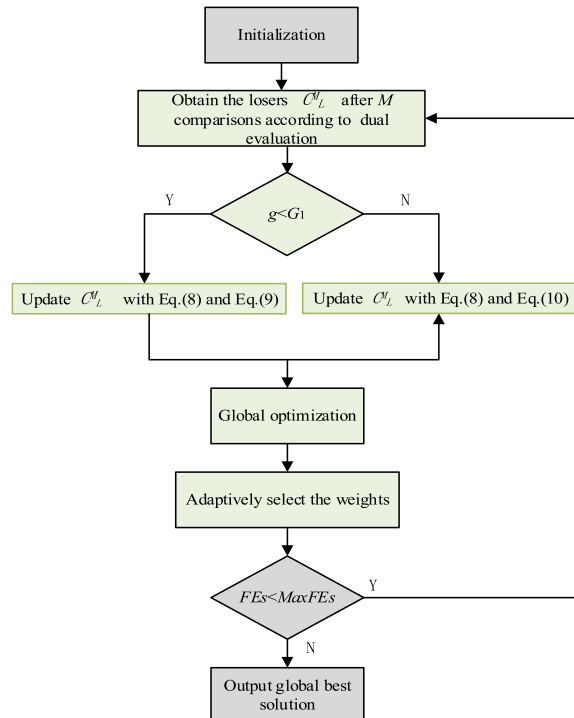


Fig. 2. Flow chart of DEGMCSO algorithm.

new particle. The fitness values of the particle are computed, after that, we compare with the original globally optimal fitness value. If it is better than original particle, it will become new the best individual, and finally the better particle is obtained. The best individual can constantly self-update modify based on global optimization.

DEGMCSO is shown in Algorithm 1. Fig.2 demonstrates flow chart of DEGMCSO.

Algorithm 1 The pseudo code of DEGMCSO.

Begin

01. $FES = 0, \varphi_1 = 0.3$;
 02. Create an initial population $x_i^G, i = 1, 2 \dots NP$ within the solution space;
 03. Set initial weight value $\omega_0 = 0.5$;
 04. Evaluate the initial population based on fitness;
 05. Calculate the score based on weighted fitness-distance;
 06. $FES = FES + NP$;
 07. **While** $FES \leq MaxFES$
 08. Choose two particles from the current population randomly;
 09. Generate the winners C_w^1 and the losers C_L^1 by the first competition according to the fitness value;
 10. Generate the losers C_L^2 and winners C_w^2 from the winner C_w^1 through the second competition according to score based on the score of weighted fitness-distance. Obtain the losers and winners from the loser group through M competitions based on the score of the weight fitness and distance evaluation, the losers after M competitions is recorded as C_L^M ;
 11. **If** $FES < FES_1$
 12. Update C_L^M by Eq.(8) and Eq. (9)
 13. **Else**
 14. Update the C_L^M by (8) and (10)
 15. **End**
 16. Calculate the fitness of C_L^M ;
 17. Calculate the score based on weighted fitness-distance of C_L^M ;
 18. $FES = FES + \lceil \frac{NP}{2^{M-1}} \rceil$;
 19. Execute global optimization and obtain new candidate global best solution
 20. Calculate the fitness of new candidate global best solution
 21. $FES = FES + 1$
 22. Update global best solution
 23. Calculate R_i by Eq.(11)
 24. Determine ω in next cycle
 25. **End while**
 26. **End**
-

4. Empirical analysis of DEGMCSO

To verify the feasibility and validity of DEGMCSO, the experiments are conducted on CEC 2010 test suit and CEC 2013 test suit. Since CEC 2013 introduces more complex properties, such as imbalances and overlaps, the optimization problems in CEC 2013 are more difficult than those in the set of CEC 2010. With the introduction of pathology, irregularity and symmetry breaking, CEC2013 simulates real-world large-scale problems. A series of comparisons are implemented between DEGMCSO and some popular methods for LSOPs, such as SLPSO, CSO, CCPSO2, TPLSO, DLLSO, MSORL, MCSO, DECC-DG [30], SDLSO [31], RLLPSO [32] and SinLSO [33].

4.1. Parameter analysis for DEGMCSO

To analyze the algorithm performance of different parameter settings, CEC2013 test set is used to conduct some experiments, here we mainly analyze two key parameters: the number of competitions M , control factor φ_1 and population size NP . Fig. 3 displays the results. The left side shows number of wins, draws and losses for the current parameter on the CEC2013 data set compared to other parameter settings, and each row on the right represents the number of each function compared to other parameter settings.

For the number M of the comparisons, M is selected from the set $S_M = [3, 4, 5, 6]$. It is obvious that $M = 4$ gets better performance than other settings in set S_M from Fig. 3. When the number of comparisons is too few or too many, DEGMCSO's performance will be weakened. The number of comparison has great influence on DEGMCSO's optimization performance. DEGMCSO achieves the best performance at about 0.3 for the control factor φ_1 . DEGMCSO is sensitive to the impact of control factor φ_1 . The values of $\varphi_1 = 0.2$ and 0.3 usually perform than $\varphi_1 = 0.1$ as shown in Fig. 3. It is found that population size NP is critical for good performance, and the best results are obtained at $NP = 600$. When the population size is greater than 700, the performance of DEGMCSO declines. The influence for the parameters corresponds to the combination of parameters related to their sums, and the optimal performance of DEGMCSO is achieved when $M = 4, \varphi_1 = 0.3$ and $NP = 600$.

4.2. Parameter settings

The maximum value of fitness evaluations in each algorithm is set to 3,000,000. Each algorithm is performed independently at 30 times, and then the average value (mean) and standard deviations (*Std*) are used to evaluate their performance in 30 independent

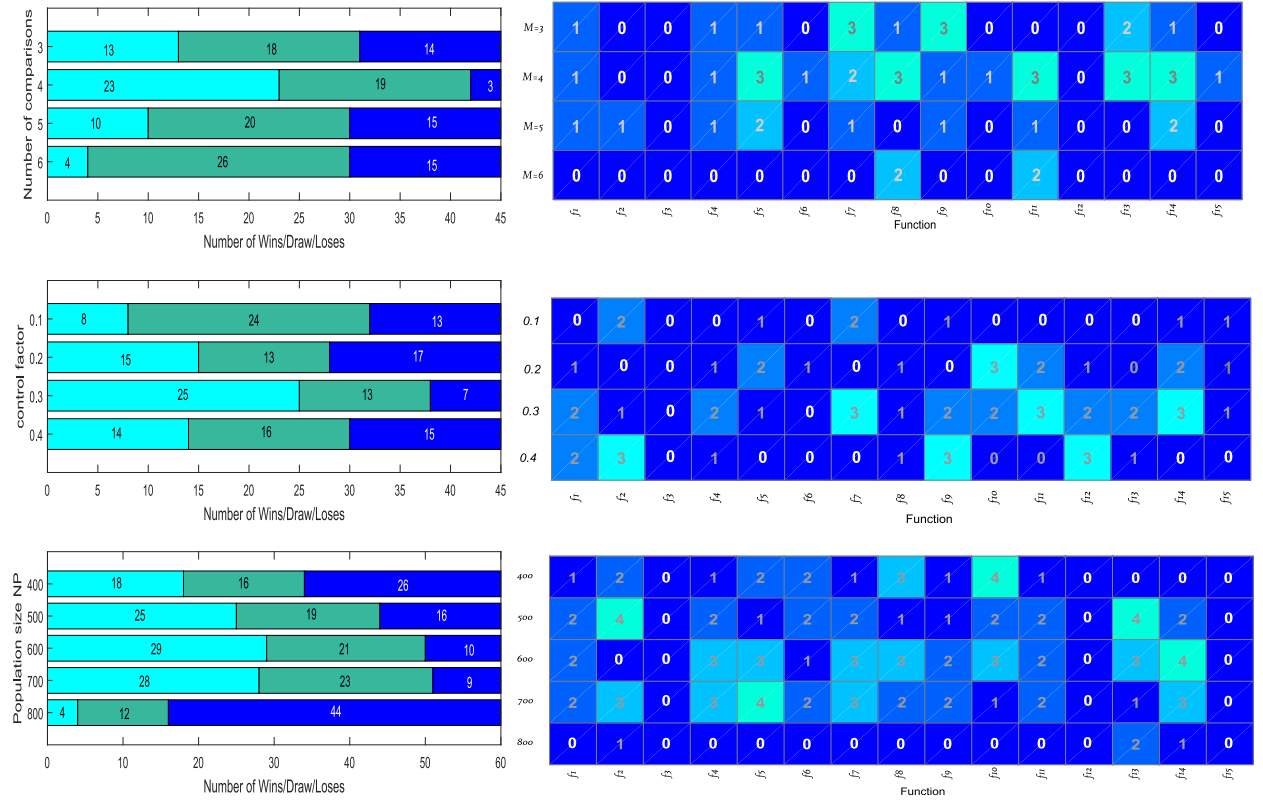


Fig. 3. The comparisons of DEGMCSO's important parameters.

executions. The parameter settings of some comparison methods are recorded in Table 1. Table 2 and Table 3 show the results of optimization experiments of DEGMCSO and other comparison algorithms on test functions of CEC 2010 and CEC 2013 test suits ($D = 1000$). CEC' 2010 [34] and CEC' 2013 [35] are developed by Li for LSOPs model test set. These two test function sets are also the most classic test function package on LSGO issued by the International Conference on Evolutionary Computing so far, and it is widely used. Fig. 4 and Fig. 7 display convergence curves of several comparison algorithms in CEC 2010 and CEC 2013 test suits. To intuitively compare DEGMCSO, Friedman test at the significance level $\alpha = 0.05$ is to rank compared algorithms. The +, – and = in last line of Table 3 and 4 obtained by Wilcoxon rank test indicates that the performance of DEGMCSO is better than, worse than or equal to the corresponding algorithm.

4.3. The comparison of CEC2010 test suite

CEC2010 large-scale optimization algorithm test set contains 20 test functions: separable functions (f_1 to f_3), partially separable functions (f_4 to f_{18}) and non-separable functions (f_{19} to f_{20}). These test functions are utilized to verify the optimization ability of DEGMCSO algorithm in handling 1000- D LSOPs. In addition to the above several methods, we compared DEGMCSO with PSO variants

Table 1
Parameters settings.

Algorithm	Parameters settings
DEGMCSO	$NP = 600, \varphi_1 = 0.3, S = \{0.1, 0.2, 0.5, 0.8\}, M = 4$
CSO	$NP = 600, \varphi_1 = 0.1, \varphi_2 = 0.15$
SLPSO	$NP = 200, M = 100, \beta = 0.001, \varphi_2 = 0.15$
CCPSO2	$NP = 30, S = \{2, 5, 10, 50, 100, 250\}$
MCSO	$NP = 400, \varphi_1 = 0.1, \varphi_2 = 0.2$
DECC-DG	$NP = 50, \varepsilon = 1e-3$
RLLPSO	$NP = 500, \varphi = 0.4$
SDLSO	$NP = 400$
TPLSO	$NP = 600, \varphi_1 = 0.15, \varphi_2 = 0.2, K = 3$
MSORL	$NP = 40, NS = 20, \varphi_1 = 0.4, \varphi_2 = 0.4, \alpha = 0.4, \gamma = 0.4$
SinSLSO	$N_{max} = 800, N_{min} = 400, \varphi_{max} = 0.3, \varphi_{min} = 0.25, NFE_{max} = 3 \times 10^6, NFE_{tp} = NFE_{max}/3$
DLLSO	$NP = 500, \varphi = 0.4, S = \{4, 6, 8, 10, 20, 50\}$

Table 2

The experiment results on 1000D test functions of CEC2013 (Mean/Std).

Fun	DEGMCSO	CSO	SLPSO	CCPSO2	MCSO	DECC-DG
f_1	0.00E + 00/0.00E + 00	4.19E-12/5.58E-13(+)	3.69E-18/1.91E-18 (+)	1.97E + 00/4.26E + 00(+)	4.97E-18/3.50E-19 (+)	1.29E + 00/2.36E-01(+)
f_2	1.74E + 02/3.04E + 01	7.60E + 03/3.06E + 02(+)	1.87E + 03/2.78E + 02(+)	6.32E + 02/5.39E + 00(+)	1.05E + 03/4.31E + 02(+)	4.37E + 03/4.59E + 01(+)
f_3	1.42E-14/3.07E-16	2.58E-09/3.47E-10(+)	2.06E + 00/3.95E-01 (+)	4.39E-03/9.67E-04 (+)	5.02E-13/5.04E-14 (+)	1.70E + 01/4.75E-01(+)
f_4	5.26E + 10/4.73E + 09	1.30E + 12/2.23E + 11(+)	4.12E + 11/8.59E + 10(+)	4.02E + 12/7.95E + 11(+)	1.20E + 12/5.79E + 11(+)	5.39E + 12/1.71E + 12(+)
f_5	4.98E + 06/3.20E + 05	5.69E + 06/9.73E + 05(+)	2.20E + 07/5.96E + 06(+)	5.99 E + 08/7.98E + 07(+)	1.50E + 07/8.31E + 06(+)	1.29E + 08/1.93E + 07(+)
f_6	3.62E-10/2.03E-13	8.01E-07/7.26E-08(+)	1.99E + 01/5.83E + 00(+)	2.01E + 07/4.93E + 06(+)	4.00E-07/4.32E-08 (+)	1.68E + 01/4.07E-01(+)
f_7	1.24E-10/2.56E-11	1.88E + 04/3.82E + 03(+)	4.19E + 04/5.59E + 03(+)	1.79E + 10/8.58E + 09(+)	6.49E + 02/9.28E + 01(+)	1.10E + 04/5.62E + 03(+)
f_8	5.39E + 03/6.09E + 02	3.79E + 07/8.17E + 05(+)	1.40E + 07/6.82E + 06(+)	8.39E + 07/4.01E + 06(+)	2.98E + 07/3.05E + 06(+)	4.41E + 07/8.79E + 06(+)
f_9	6.67E + 06/1.22E + 06	5.88E + 07/5.50E + 06(+)	3.40E + 07/3.40E + 07(+)	1.97E + 08/6.15E + 07(+)	4.29E + 07/9.83E + 06(+)	4.39E + 07/4.15 E + 06(+)
f_{10}	7.05E + 02/3.14E + 01	9.59E + 03/7.21E + 01(+)	3.11E + 03/4.25E + 02(+)	6.01E + 03/8.39E + 02(+)	1.08E + 03/5.98E + 02(+)	4.51E + 03/4.84E + 02(+)
f_{11}	9.23E-14/2.15E-15	4.38E-08/3.54E-09(+)	2.19E + 01/2.04E + 00(+)	2.00E + 02/2.31E + 00(+)	5.49E-12/4.03E-13 (+)	1.11E + 01/8.13E-01(+)
f_{12}	7.75E + 01/1.27E + 01	5.50E + 05/7.33E + 04(+)	1.69E + 04/7.38E + 03(+)	1.69E + 05/2.89E + 04(+)	2.18E + 04/9.67E + 03(+)	1.22E + 03/1.15E + 02(+)
f_{13}	3.01E + 02/7.83E + 01	6.97E + 02/9.74E + 01(+)	8.56E + 02/8.05E + 01(+)	2.54E + 03/2.03E + 02(+)	2.23E + 03/6.28E + 02(+)	3.89E + 03/4.03E + 02(+)
f_{14}	1.56E + 07/3.02E + 06	2.75E + 08/8.94E + 06(+)	1.20E + 08/6.09E + 07(+)	4.29E + 08/9.33E + 07(+)	1.29E + 08/5.85E + 07(+)	3.50E + 08/9.61E + 07(+)
f_{15}	2.13E + 03/5.26E + 02	9.98E + 03/6.32E + 01(+)	1.10E + 04/3.01E + 03(+)	1.40E + 04/2.37E + 03(+)	7.60E + 02/1.62E + 02(-)	6.01E + 03/5.23E + 02(+)
f_{16}	9.94E-14/1.77E-15	6.01E-08/2.40E-09(+)	3.70E + 01/6.58E + 00(+)	3.98E + 02/8.31E + 00(+)	1.50E + 00/4.73E-01 (+)	5.79E-13/7.91E-14 (+)
f_{17}	1.85E + 03/1.57E + 03	2.31E + 06/1.03E + 05(+)	8.97E + 04/9.61E + 03(+)	4.20E + 05/9.68E + 04(+)	1.09E + 05/5.81E + 04(+)	3.90E + 04/4.86E + 03(+)
f_{18}	7.72E + 02/4.66E + 01	2.14E + 03/3.08E + 02(+)	3.48E + 03/7.39E + 02(+)	2.31E + 04/6.39E + 03(+)	3.81E + 03/5.19E + 02(+)	1.47E + 10/3.01E + 09(+)
f_{19}	9.94E + 05/4.54E + 04	1.09E + 07/2.68E + 06(+)	5.80E + 06/6.68E + 05(+)	2.78E + 06/8.64E + 05(+)	2.99E + 06/4.26E + 05(+)	1.66E + 06/8.59E + 04(+)
f_{20}	9.04E + 02/2.37E + 02	1.05E + 03/4.30E + 02(+)	1.59E + 03/4.21E + 02(+)	1.80E + 03/4.01E + 02(+)	2.11E + 03/3.82E + 02(+)	6.49E + 10/4.84E + 09(+)
+/-/-rank	2.35	20/0/0	20/0/0	20/0/0	19/0/1	20/0/0
Fun	RLLPSO	TPLSO	SDLPO	MSORL	SinSLSO	DLLSO
f_1	4.49E-22/8.76E-23 (+)	2.98E-19/3.14E-20(+)	8.29E-20/6.86E-21 (+)	8.01E-22/1.21E-22 (+)	1.18E-21/1.38E-22 (+)	4.40E-22/6.15E-23 (+)
f_2	9.79E + 02/5.85E + 01(+)	1.08E + 03/5.51E + 02(+)	4.98E + 02/3.79E + 01(+)	6.66E + 02/2.74E + 01(+)	7.61E + 02/4.20E + 01(+)	9.79E + 02/4.03E + 01(+)
f_3	8.69E-14/9.25E-15 (+)	9.09E-14/3.12E-15(+)	2.60E-13/3.04E-14 (+)	3.47E-14/3.82E-14 (+)	4.79E-13/3.86E-14 (+)	2.84E-14/3.05E-15 (+)
f_4	6.50E + 11/4.93E + 10(+)	3.80E + 11/5.03E + 10(+)	5.01E + 11/6.78E + 10(+)	1.40E + 11/3.19E + 10(+)	3.16E + 11/5.60E + 10(+)	4.48E + 11/7.76E + 10(+)
f_5	1.29E + 07/4.01 E + 06(+)	1.60E + 07/5.82E + 06(+)	7.71E + 06/7.92E + 05(+)	5.57E + 06/1.68E + 06(+)	1.09E + 07/5.21E + 06(+)	1.18E + 07/5.04E + 06(+)
f_6	3.38E-01/6.04E-02 (+)	1.79E-07/2.40E-08(+)	9.08E-09/9.84E-10 (+)	1.20E + 00/4.69E-01 (+)	3.60E-09/6.86E-13 (+)	5.41E-01/8.73E-02 (+)
f_7	3.10E-01/1.97E-01 (+)	7.51E + 03/9.36E + 02(+)	4.79E + 02/8.67E + 01(+)	5.89E-04/3.47E-04 (+)	1.06E-01/8.79E-02 (+)	7.20E + 02/1.30E + 02(+)
f_8	1.39E + 07/8.65E + 06(+)	3.79E + 06/4.83E + 05(+)	3.30E + 07/2.01E + 06(+)	2.38E + 04/1.05E + 04(+)	8.97E + 05/6.13E + 04(+)	2.91E + 07/4.05E + 05(+)
f_9	3.09E + 07/2.05E + 06(+)	2.50E + 07/3.29E + 06(+)	5.62E + 07/4.15E + 06(+)	1.28E + 07/1.06E + 06(+)	2.40E + 07/3.05E + 06(+)	4.18E + 07/5.83E + 06(+)
f_{10}	9.09E + 02/3.35E + 01(+)	7.50E + 02/4.28E + 01(+)	6.39E + 03/3.94E + 02(+)	5.72E + 02/3.14E + 01(-)	7.29E + 02/4.98E + 01(+)	8.79E + 02/4.39E + 01(+)
f_{11}	2.49E + 00/4.98 E-01 (+)	3.38E + 00/1.34E + 00(+)	2.23E + 01/2.45E + 00(+)	1.86E + 01/4.07E + 00(+)	2.10E-13/3.87E-14 (+)	4.90E + 00/9.77E-01(+)
f_{12}	1.20E + 04/7.31E + 02(+)	6.50E + 03/9.49E + 02(+)	8.89E + 04/8.54E + 03(+)	5.49E + 03/3.85E + 02(+)	1.87E + 04/4.01E + 03(+)	1.41E + 04/5.69E + 03(+)

(continued on next page)

Table 2 (continued)

Fun	DEGMCSO	CSO	SLPSO	CCPSO2	MCSO	DECC-DG
f_{13}	9.69E + 02/9.51E + 01(+)	8.29E + 02/9.38E + 01(+)	6.40E + 02/7.96E + 01(+)	5.51E + 02/1.04E + 02(+)	6.29E + 02/9.67E + 01(+)	9.38E + 02/7.52E + 01(+)
f_{14}	1.08E + 08/4.27E + 06(+)	1.09E + 08/2.15E + 07(+)	1.21E + 08/2.61E + 07(+)	4.01E + 07/2.90E + 06(+)	7.39E + 07/5.83E + 06(+)	1.30E + 08/1.09E + 07(+)
f_{15}	8.39E + 02/6.31E + 01(−)	9.98E + 03/5.96E + 01(+)	9.59E + 03/3.04E + 02(+)	9.79E + 03/5.13E + 01(+)	8.59E + 02/5.04E + 01(−)	8.49E + 02/5.03E + 01(−)
f_{16}	1.87E + 00/8.76E-01(+)	2.10E-12/4.15E-13(+)	5.20E-12/5.40E-13(+)	4.49E + 00/2.36E + 00(+)	3.11E-13/2.54E-14(+)	2.39E + 00/9.84E-01(+)
f_{17}	5.39E + 04/2.04E + 03(+)	8.59E + 04/7.64E + 03(+)	7.40E + 05/4.03E + 04(+)	1.95E + 05/1.71E + 04(+)	1.71E + 05/3.18E + 04(+)	5.30E + 04/3.61E + 03(+)
f_{18}	2.12E + 03/4.03E + 02(+)	2.20E + 03/5.10E + 02(+)	1.69E + 03/5.22E + 02(+)	1.40E + 03/3.31E + 02(+)	1.12E + 03/4.58E + 02(+)	2.40E + 03/7.96E + 02(+)
f_{19}	8.75E + 05/3.96E + 04(+)	3.40E + 06/4.18E + 05(+)	8.70E + 06/6.45E + 05(+)	6.62E + 06/3.39E + 05(+)	5.28E + 06/5.73E + 05(+)	1.79E + 06/2.74E + 05(+)
f_{20}	1.89E + 03/3.29E + 02(+)	1.51E + 03/2.34E + 02(+)	1.77E + 03/3.14E + 02(+)	1.19E + 03/8.26E + 01(+)	9.49E + 02/3.65E + 01(+)	1.80E + 03/3.06E + 02(+)
+/-/-rank	19/0/1 7.70	20/0/0 8.45	20/0/0 10.15	19/0/1 6.60	19/0/1 6.05	19/0/1 8.50
Fun	MOS	MLSHADE-SPA	MA-SW-Chains	CCGS	L-SHADE	FDB-AGDE
f_1	0.00E + 00/0.00E + 00(=)	4.01E-23/2.88E-23(+)	2.09E-14/1.87E-14(+)	1.83E-22/3.68E-22(+)	1.83E + 09/2.01E + 08(+)	3.48E-21/4.59E-22(+)
f_2	2.01 E + 02/3.05E + 01(+)	4.69E + 01/7.83E + 00(−)	8.09E + 02/6.03E + 01(+)	4.38E-02/2.29E-01(−)	3.92E + 03/4.78E + 02(+)	1.12E + 03/5.71E + 02(+)
f_3	1.10E + 00/8.94E-01(+)	9.79E-14/8.59E-15(+)	7.30E-13/4.05E-13(+)	1.89E-01/4.49E-01(+)	1.18E + 01/3.65E + 00(+)	7.10E-14/3.87E-15(+)
f_4	2.02E + 10/7.63E + 09(−)	1.84E + 11/5.72E + 10(+)	3.49E + 11/3.30E + 10(+)	1.79E + 12/7.59E + 11(+)	3.03E + 10/5.08E + 09(−)	6.79E + 11/7.60E + 10(+)
f_5	7.01E + 08/9.65E + 07(+)	6.10E + 07/8.61E + 06(+)	1.70E + 08/2.13E + 08(+)	1.97E + 07/4.69E + 06(+)	1.58E + 04/2.13E + 03(−)	1.84E + 08/1.39E + 08(+)
f_6	2.01E + 07/7.12E + 04(+)	6.58E-09/9.64E-08(+)	8.09E + 04/3.76E + 04(+)	2.88E + 06/5.03E + 05(+)	2.11E + 01/5.45E + 00(+)	1.02E + 00/6.02E-01(+)
f_7	0.00E + 00/0.00E + 00(−)	8.07E-06/4.53E-07(+)	1.04E + 02/6.34E + 01(+)	1.36E + 02/9.98E + 01(+)	2.33E + 06/5.09E + 05(+)	3.45E + 02/5.40E + 01(+)
f_8	1.22E + 06/1.05E + 06(+)	4.39E + 03/3.02E + 03(−)	1.39E + 07/2.95E + 07(+)	2.81E + 07/3.14E + 07(+)	2.95E + 07/4.77E + 06(+)	3.58E + 07/2.01E + 05(+)
f_9	8.90E + 06/8.74E + 05(+)	1.64E + 07/3.11E + 06(+)	1.42E + 07/3.05E + 06(+)	5.53E + 07/9.60E + 06(+)	1.92E + 09/3.96E + 08(+)	6.12E + 07/4.53E + 06(+)
f_{10}	7.80E + 03/4.01E + 02(+)	4.98E + 03/3.04E + 02(+)	2.08E + 03/2.01E + 02(+)	4.74E + 03/2.45E + 03(+)	4.33E + 03/5.78E + 02(+)	4.48E + 03/3.10E + 02(+)
f_{11}	1.86E + 02/4.43E-01(+)	8.58E + 01/9.54E + 01(+)	3.79E + 01/6.48E + 00(+)	3.00E + 01/3.71E + 00(+)	8.51E + 01/6.04E + 00(+)	6.52E + 01/4.76E + 00(+)
f_{12}	0.00E + 00/0.00E + 00(−)	4.29E + 01/8.37E + 00(−)	3.59E-06/6.15E-07(−)	5.35E + 03/5.07E + 02(+)	7.07E + 05/5.35E + 04(+)	4.12E + 04/4.02E + 03(+)
f_{13}	1.39E + 03/7.85E + 02(+)	9.97E + 01/8.61E + 00(−)	1.26E + 03/4.87E + 02(+)	1.51E + 03/6.94E + 02(+)	8.12E + 06/3.67E + 05(+)	1.31E + 03/5.37E + 02(+)
f_{14}	2.01E + 07/2.41E + 06(+)	5.98E + 07/5.79E + 06(+)	3.09E + 07/1.87E + 06(+)	1.35E + 08/9.05E + 06(+)	2.04E + 09/5.71E + 08(+)	2.11E + 08/2.51E + 07(+)
f_{15}	1.58E + 04/6.13E + 02(+)	3.81E + 03/8.70E + 01(+)	2.74E + 03/3.01E + 02(+)	1.74E + 03/8.94E + 01(−)	4.41E + 03/5.81E + 02(−)	6.24E + 03/4.16E + 02(+)
f_{16}	3.99E + 02/3.28E-01(+)	2.50E + 02/2.05E + 01(+)	9.91E + 01/8.76E + 00(+)	3.11E + 01/5.22E + 00(+)	2.29E + 02/4.29E + 01(+)	3.29E + 02/3.47E + 01(+)
f_{17}	4.68E-05/1.97E + 03(−)	6.16E + 02/9.20E + 01(−)	1.22E + 00/2.85E-01(−)	1.48E + 04/1.45E + 03(+)	7.94E + 05/3.55E + 04(+)	3.35E + 05/5.28E + 04(+)
f_{18}	4.02E + 03/2.57E + 03(+)	5.73E + 02/1.76E + 02(−)	1.30E + 03/4.36E + 02(+)	3.13E + 03/1.01E + 03(+)	7.33E + 09/2.61E + 08(+)	1.75E + 03/6.56E + 02(+)
f_{19}	3.56E + 04/2.90E + 03(−)	4.51E + 05/2.34E + 04(−)	2.90E + 05/1.67E + 04(−)	5.93E + 05/4.05E + 04(−)	1.06E + 06/4.79E + 05(+)	2.23E + 06/4.09E + 05(+)
f_{20}	8.43E + 02/3.89E + 02(−)	1.26E + 02/9.55E + 01(−)	1.08E + 03/8.09E + 01(+)	1.31E + 03/2.14E + 02(+)	7.10E + 09/3.68E + 08(+)	1.86E + 03/3.14E + 02(+)
+/-/-rank	13/1/6 8.45	12/0/8 5.55	17/0/3 9.40	17/0/3 9.70	18/0/2 13.60	20/0/0 12.10

with some better performance proposed in recent years. The compared algorithms include: the winner of the CEC2010, MA-SW-Chains and the top four algorithms of MOS, MLSHADE-SPA [36] and CCGS [37], L-SHADE [38] and FDB-AGDE [39].

Table 2 records DEGMCSO and other 17 algorithms the test results of CEC '2010 set. To increase the credibility, the parameter settings of the other 17 algorithms are consistent with those used in the paper. For the clarity, each of the best results is shown in bold font. The result of t test indicates that DEGMCSO has a good advantage in statistical optimization ability compared with all of comparison algorithms.

Table 3

The experiment results on 1000D test functions of CEC2013 (Mean/Std).

Fun	DEGMCSSO	CSO	SLPSO	CCPSO2	MCSSO	DECC-DG
f_1	0.00E + 00/0.00E + 00	7.20E-12/3.91E-13(+)	1.51E-17/3.28E-18 (+)	5.23E + 00/5.03E + 00(+)	5.78E-18/3.91E-19 (+)	1.37E + 02/4.84E + 01(+)
f_2	1.64E + 02/4.68E + 01	8.49E + 03/3.15E + 02(+)	2.09E + 03/9.54E + 02(+)	1.80E + 02/9.25E + 01(+)	1.39E + 03/4.76E + 02(+)	1.28E + 04/8.70E + 03(+)
f_3	2.00E + 01/3.40E-04	2.16E + 01/5.10E-03 (+)	2.16E + 01/1.94E-02(+)	2.00E + 01/2.37E-04 (=)	2.16E + 01/2.04E-03 (+)	2.15E + 01/3.04E-02(+)
f_4	1.37E + 08/6.35E + 07	1.60E + 10/2.79E + 09(+)	6.31E + 09/8.95E + 08(+)	3.90E + 10/9.87E + 09(+)	7.58E + 09/8.96E + 08(+)	4.29E + 10/7.60E + 09(+)
f_5	4.18E + 05/4.73E + 04	1.51E + 06/5.18E + 05(+)	8.05E + 05/9.40E + 04(+)	1.92E + 07/5.03E + 06(+)	7.19E + 05/4.83E + 04(+)	5.86E + 06/4.05E + 05(+)
f_6	1.02E + 06/8.05E + 02	1.06E + 06/2.05E + 03(+)	1.06E + 06/2.57E + 03(+)	1.06E + 06/6.87E + 03(+)	1.06E + 06/6.05E + 03(+)	1.06E + 06/9.80E + 02(+)
f_7	1.11E + 05/6.73E + 04	4.01E + 06/8.57E + 05(+)	1.51E + 06/6.89E + 05(+)	8.29E + 08/7.05E + 07(+)	1.90E + 06/6.87E + 08(+)	4.68E + 08/3.65E + 07(+)
f_8	7.84E + 12/1.35E + 12	2.58E + 14/4.05E + 13(+)	9.28E + 13/9.65E + 12(+)	2.49E + 15/5.17E + 14(+)	1.59E + 14/9.69E + 14(+)	4.28E + 15/8.70E + 14(+)
f_9	2.57E + 07/1.83E + 06	5.79E + 07/7.18E + 06(+)	8.77E + 07/2.63E + 07(+)	2.97E + 09/1.04E + 09(+)	4.19E + 07/2.84E + 07(+)	4.76E + 08/2.32E + 07(+)
f_{10}	9.09E + 07/5.85E + 04	9.40E + 07/1.56E + 05(+)	9.37E + 07/1.77E + 06(+)	9.29E + 07/5.04E + 05(+)	9.43E + 07/1.03E + 06(+)	9.46E + 07/3.29E + 05(+)
f_{11}	1.94E + 06/9.15E + 05	1.85E + 09/1.02E + 09(+)	4.28E + 10/7.65E + 10(+)	9.36E + 11/3.45E + 10(+)	1.88E + 08/7.49E + 07(+)	1.10E + 10/2.46E + 10(+)
f_{12}	9.31E + 02/6.03E + 01	1.07E + 03/2.89E + 01(+)	1.83E + 03/7.54E + 02(+)	1.69E + 03/2.85E + 02(+)	1.91E + 03/5.37E + 02(+)	1.66E + 11/2.30E + 10(+)
f_{13}	2.22E + 07/1.39E + 07	7.68E + 08/3.49E + 08(+)	4.65E + 08/2.31E + 08(+)	7.03E + 09/1.68E + 09(+)	5.80E + 08/3.41E + 08(+)	1.53E + 10/5.86E + 09(+)
f_{14}	5.51E + 06/3.84E + 05	5.14E + 09/1.05E + 08(+)	6.79E + 08/5.47E + 08(+)	9.33E + 10/8.79E + 10(+)	1.37E + 09/1.43E + 08(+)	2.91E + 10/1.42E + 10(+)
f_{15}	1.36E + 06/1.27E + 05	7.34E + 07/2.30E + 06(+)	6.19E + 07/6.52E + 06(+)	5.42E + 06/5.09E + 06(+)	5.48E + 06/3.72E + 06(+)	1.00E + 07/2.35E + 06(+)
+/-/0		15/0/0	15/0/0	14/1/0	15/0/0	15/0/0
rank	2.67	11.73	10.87	12.87	10.93	15.33
Fun	RLLPSO	TPLSO	SDLSO	MSORL	SinSLSO	DLLSO
f_1	9.20E-22/8.78E-23(+)	3.30E-19/4.85E-20(+)	4.30E-23/1.27E-23 (+)	1.04E-21/2.21E-22 (+)	1.37E-21/3.10E-22 (+)	4.59E-22/3.78E-23 (+)
f_2	1.05E + 03/9.20E + 01(+)	7.19E + 02/9.35E + 02(+)	8.29E + 02/5.37E + 01(+)	8.79E + 02/4.70E + 01(+)	7.88E + 02/4.69E + 01(+)	1.30E + 03/6.01E + 01(+)
f_3	2.09E + 01/7.29E-02 (+)	2.16E + 01/5.74E-01 (+)	2.16E + 01/7.09E-03(+)	2.16E + 01/6.71E-03 (+)	2.16E + 01/5.84E-03 (+)	2.16E + 01/4.19E-03(+)
f_4	4.41E + 09/4.67E + 08(+)	5.59E + 09/4.61E + 08(+)	4.10E + 09/6.72E + 08(+)	1.31E + 09/3.67E + 08(+)	3.19E + 09/4.91 E + 08(+)	6.99E + 09/7.85E + 08(+)
f_5	7.29E + 05/1.31E + 05(+)	6.61E + 05/7.23E + 04(+)	6.23E + 05/8.63E + 04(+)	5.20E + 05/6.91E + 04(+)	5.39E + 05/6.85E + 04(+)	7.31E + 05/1.06E + 05(+)
f_6	1.06E + 06/8.94E + 02(+)	1.06E + 06/8.71E-03 (+)	1.06E + 06/9.58E + 02(+)	1.06E + 06/8.23E + 02(+)	1.06E + 06/2.67E + 03(+)	1.06E + 06/8.30E + 02(+)
f_7	9.91E + 05/8.79E + 04(+)	1.10E + 06/4.84E + 05(+)	1.21E + 06/8.89E + 05(+)	3.37E + 05/1.62E + 05(+)	1.49E + 05/3.08E + 05(+)	1.70E + 06/7.54E + 05(+)
f_8	8.52E + 13/9.47E + 13(+)	1.40E + 14/3.89E + 13(+)	1.05E + 14/2.69E + 13(+)	3.72E + 13/1.23E + 13(+)	5.99 E + 13/8.95E + 12(+)	1.59E + 14/4.06E + 13(+)
f_9	3.87E + 07/2.09E + 06(+)	2.39E + 07/3.61E + 06(-)	2.03E + 08/1.20E + 08(+)	8.67E + 07/2.25E + 07(+)	3.50E + 07/6.03E + 06(+)	1.49E + 08/2.98E + 07(+)
f_{10}	9.39E + 07/4.95E + 05(+)	9.40E + 07/3.59E + 05(+)	9.41E + 07/2.11E + 05(+)	9.40E + 07/2.85E + 05(+)	9.41E + 07/4.14E + 05(+)	9.40E + 07/4.12E + 05(+)
f_{11}	9.31E + 11/8.46E + 09(+)	1.29E + 08/3.96E + 07(+)	1.07E + 09/1.33E + 09(+)	9.57E + 11/3.81E + 10(+)	7.79E + 07/9.52E + 06(+)	9.30E + 11/2.35 E + 10(+)
f_{12}	1.89E + 03/2.15E + 02(+)	1.08E + 03/3.07E + 02(+)	1.17E + 03/7.56E + 01(+)	1.32E + 03/8.92E + 01(+)	1.02E + 03/3.89E + 01(+)	1.98E + 03/3.28E + 02(+)
f_{13}	2.37E + 08/4.83E + 07(+)	9.07E + 07/8.96E + 06(+)	1.90E + 09/6.75E + 08(+)	2.09E + 08/9.30E + 07(+)	1.09E + 08/5.26E + 07(+)	4.20E + 08/7.65E + 07(+)
f_{14}	5.09E + 07/8.75E + 06(+)	2.30E + 07/4.75E + 06(+)	6.84E + 09/5.05E + 09(+)	4.69E + 07/1.07E + 07(+)	1.36E + 08/7.82E + 06(+)	1.59E + 08/8.49E + 07(+)

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Table 3 (continued)

Fun	DEGMCSO	CSO	SLPSO	CCPSO2	MCSO	DECC-DG
f_{15}	1.59E + 06/5.80E + 05(+) +/-/- Rank 8.93	2.40E + 08/3.83E + 07(+) 14/0/1 8.27	1.50E + 08/3.90E + 07(+) 15/0/0 10.67	2.39E + 07/2.46E + 06(+) 15/0/0 9.07	1.70E + 07/1.37E + 06(+) 15/0/0 8	5.01E + 06/5.02E + 05(+) 15/0/0 11.47
Fun	MOS	MLSHADE-SPA	CC-MA-ES	SHADE-ILS	L-SHADE	FDB-AGDE
f_1	0.00E + 00/0.00E + 00(=)	1.87E-22/5.65E-22(+)	5.79E-09/1.13E-09(+)	2.69E-24/3.57E-25(+)	4.39E + 09/3.75E + 08(+)	4.41E-21/2.40E-22(+)
f_2	8.45E + 02/3.23E + 01(+)	7.90E + 01/8.53E + 00(-)	1.33E + 03/4.75E + 02(+)	9.96E + 03/1.02E + 02(+)	1.24E + 04/3.51E + 03(+)	3.62E + 03/4.75E + 01(+)
f_3	9.20E-13/4.68E-14(-)	9.97E-14/8.32E-15(-)	1.51E-13/6.73E-15(-)	2.01E + 01/2.11E-02(+)	2.16E + 01/4.89E-03(+)	2.04E + 01/3.08E-03(+)
f_4	1.80E + 08/5.26E + 07(+)	6.87E + 08/9.32E + 07(+)	2.20E + 09/1.31E + 09(+)	1.48E + 08/6.57E + 07(+)	1.40E + 10/5.37E + 09(+)	1.19E + 10/2.05E + 09(+)
f_5	6.85E + 06/7.03E + 05(+)	1.79E + 06/3.10E + 05(+)	7.27E + 07/4.65E + 06(+)	1.39E + 06/6.20E + 05(+)	1.55E + 06/6.91E + 04(+)	2.06E + 06/4.64E + 05(+)
f_6	1.50E + 05/7.25E + 04(-)	1.39E + 03/9.87E + 02(-)	5.83E + 05/4.82E + 05(-)	1.02E + 06/7.94E + 04(-)	1.06E + 06/2.05E + 03(+)	1.05E + 06/8.41E + 02(+)
f_7	1.62E + 04/8.39E + 03(-)	5.29E + 04/2.05E + 04(-)	7.45E + 06/1.21E + 06(+)	7.39E + 01/6.52E + 01(-)	2.40E + 07/3.18E + 06(+)	5.69E + 07/2.50E + 06(+)
f_8	8.10E + 12/7.32E + 11(+)	9.79E + 12/6.03E + 12(+)	3.91E + 14/9.65E + 13(+)	3.17E + 11/8.80E + 10(-)	4.76E + 11/5.09E + 10(+)	1.50E + 14/3.71E + 13(+)
f_9	3.80E + 08/7.34E + 07(+)	1.60E + 08/1.84E + 07(+)	3.71E + 08/1.83E + 08(+)	1.64E + 08/2.70E + 07(+)	1.96E + 08/2.34E + 07(+)	1.91E + 08/4.82E + 07(+)
f_{10}	8.97E + 05/4.53E + 05(-)	6.55E + 02/1.87E + 02(-)	7.55E + 05/5.02E + 05(-)	9.18E + 07/6.36E + 05(+)	9.46E + 07/3.10E + 05(+)	9.42E + 07/3.12E + 05(+)
f_{11}	5.19E + 07/8.86E + 06(+)	4.01E + 07/9.01E + 06(+)	1.59E + 08/8.35E + 07(+)	5.11E + 05/2.11E + 05(-)	2.18E + 09/4.16E + 08(+)	1.77E + 10/6.54E + 09(+)
f_{12}	2.49E + 02/8.56E + 01(-)	1.03E + 02/6.91E + 01(-)	1.30E + 03/5.01E + 02(+)	6.18E + 01/4.10E + 00(-)	8.76E + 10/3.85E + 09(+)	1.68E + 03/2.07E + 02(+)
f_{13}	3.39E + 06/9.13E + 05(-)	7.19E + 07/9.89E + 06(+)	6.70E + 08/8.54E + 07(+)	1.00E + 05/7.33E + 04(-)	4.96E + 09/4.72E + 08(+)	1.90E + 09/3.15E + 08(+)
f_{14}	2.60E + 07/6.81E + 06(+)	1.60E + 07/4.31E + 06(+)	7.10E + 07/1.25E + 07(+)	5.76E + 06/4.77E + 05(+)	1.33E + 10/5.01E + 09(+)	4.77E + 09/2.91E + 08(+)
f_{15}	2.39E + 06/2.64E + 05(+)	2.80E + 07/8.51E + 06(+)	3.03E + 07/5.72E + 06(+)	6.25E + 05/8.03E + 04(-)	2.20E + 07/1.31E + 06(+)	8.21E + 06/4.15E + 06(+)
+/-/- rank	8/1/6 5	9/0/6 4.6	12/0/3 10.20	8/1/6 3.8	15/0/0 14.47	15/0/0 12.13

From Table 2, for 15 partially separable functions f_4 – f_{18} , the optimization results of DEGMCSO outperforms other 17 algorithms on 7 benchmark functions ($f_1, f_3, f_6, f_9, f_{11}, f_{14}$ and f_{16}). DEGMCSO achieved significantly better solution quality and faster convergence in two separable functions. Meanwhile, for partially separable functions, DEGMCSO achieves better quality and faster convergence on 5 functions than other 17 comparison algorithms.

But for non-separable functions f_{19} and f_{20} , it has room for improvement the accuracy. Although DEGMCSO is weaker than contrast algorithm in some functions, DEGMCSO still has better performance than all the contrast methods on the whole. The ranking result of each function in the function set is shown in the Fig. 5 and Fig. 6.

In particular, DEGMCSO performs significantly better on 20 functions than CSO algorithm. For function f_{10} , although the DEGMCSO algorithm is worse than MSORL algorithm, it is superior to other 16 algorithms. The DEGMCSO algorithm is inferior to MCSO, RLLPSO, SinSLSO, DLLSO and CCGS algorithms in dealing with function f_{15} . For the function f_2 , DEGMCSO is inferior to MLSHADE-SPA algorithm and CCGS algorithm, but it outperforms other 15 algorithms. In contrast, the rest of the comparison algorithms perform best on at most 6 function. In general, DEGMCSO performs better than other algorithms for solution accuracy.

This advantage of DEGMCSO is mainly due to the adaptive double evaluation strategy, which can select the particles with poor diversity and fitness from the population through multiple comparisons for different optimization problems, then these particles are ordered to actively learn from the elite particles that win through two competitions. Under the guidance of elite particles, the loser particles are expected to move rapidly in different directions to the optimal region. The global optimization is likely to help DEGMCSO find a high-quality optimization problem quickly.

Through the Wilcoxon rank-sum test, compared with CCPSO2, DECC-DG and MSORL based on CC framework, DEGMCSO achieves better results on 20, 20 and 19 functions respectively. Compared with CSO, SLPSO, MCSO and DLLSO, the algorithm DEGMCSO performs well on 20, 20, 19 and 19 test functions, respectively. Compared to the other three recent large-scale evolutionary algorithms (i.e. RLLPSO, TPLSO, SinSLSO and SDLISO) significantly outperformed them on 19, 20, 19 and 20 functions. Compared to MA-SW-Chains, DEGMCSO outperforms MA-SW-Chains on 12 functions, especially on f_6, f_7, f_8, f_{11} and f_{16} . DEGMCSO is superior to CCGS on

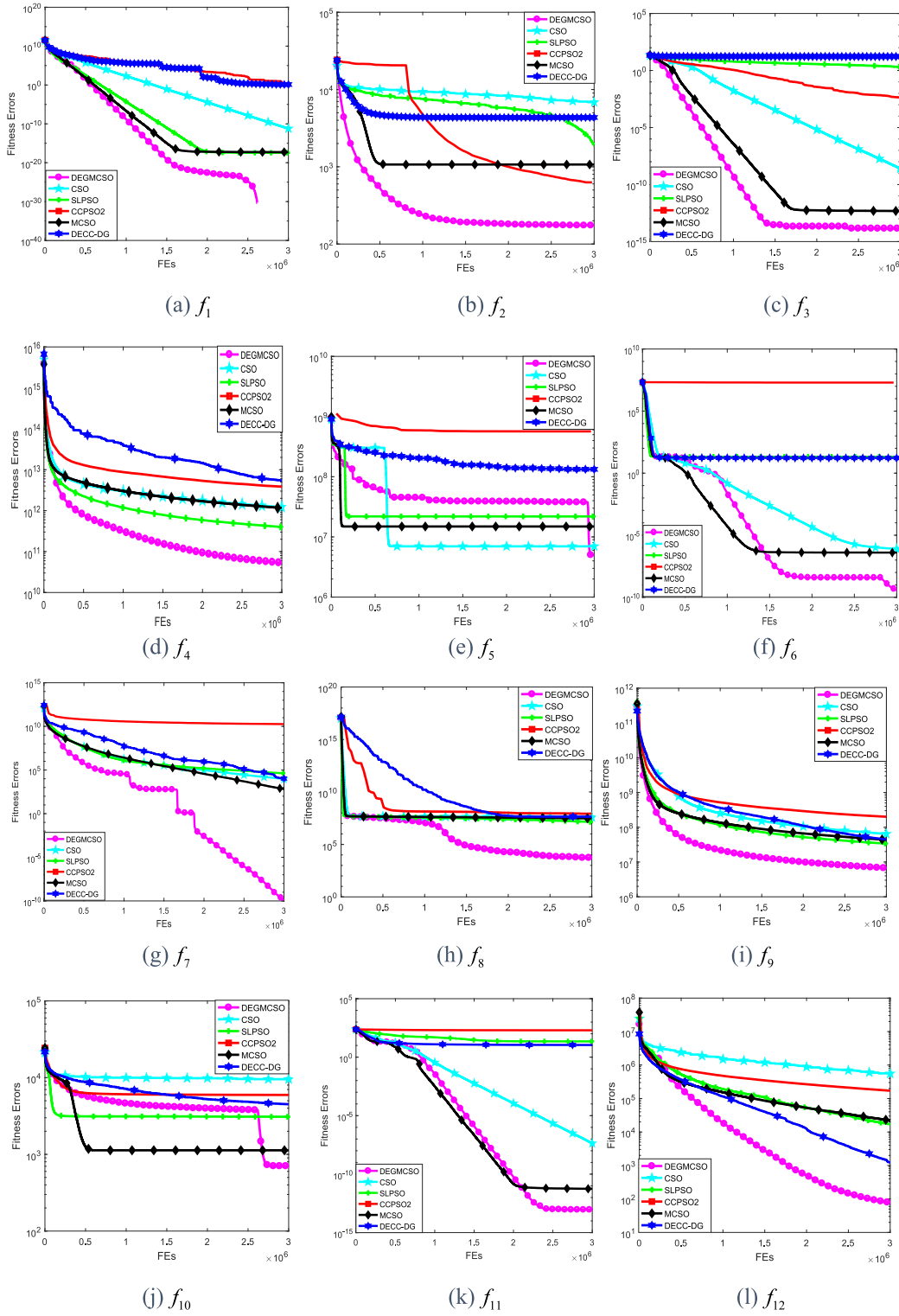


Fig. 4. Convergence curves in CEC2010 test suite.

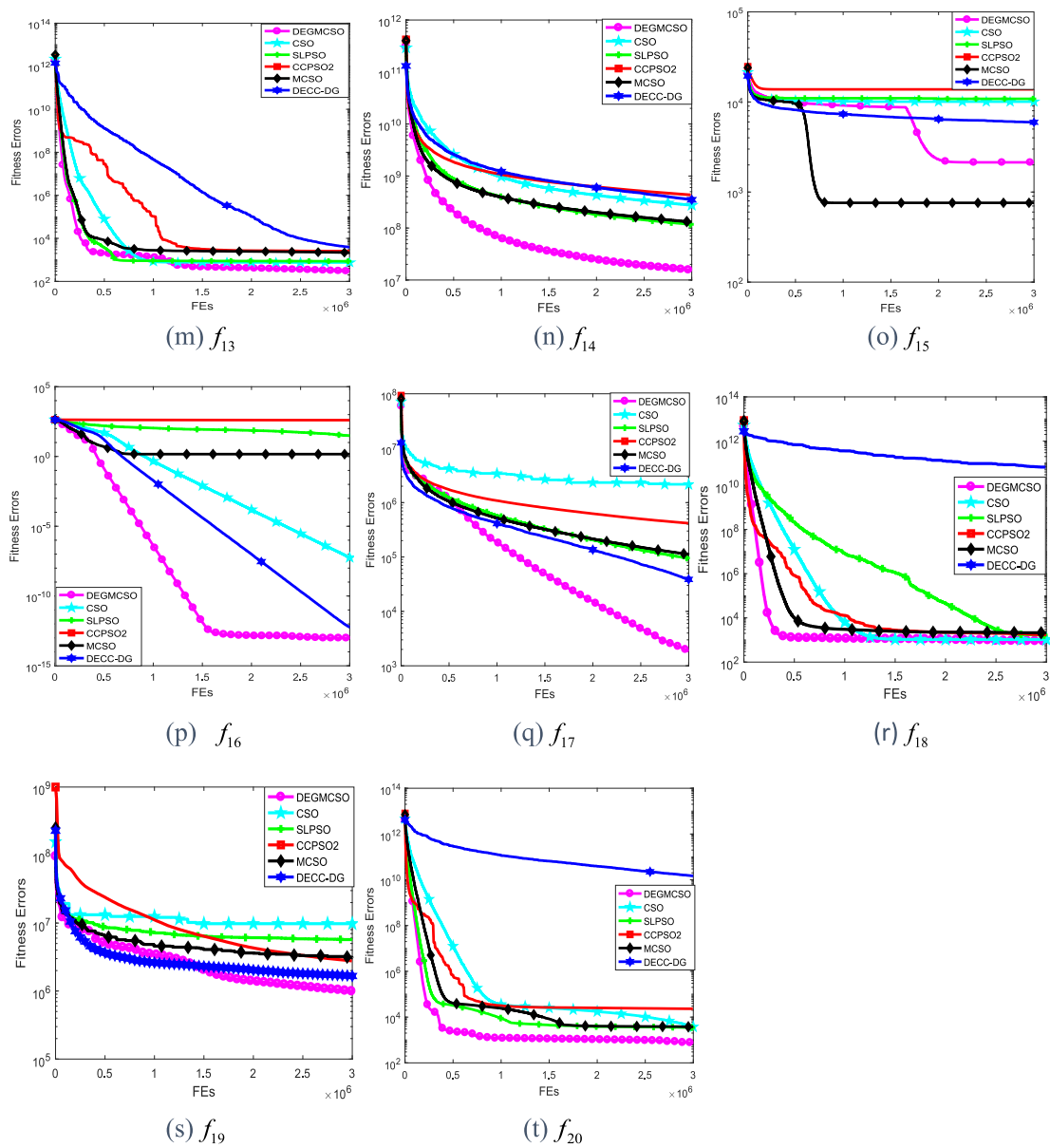


Fig. 4. (continued).

13 partially separable functions. Compared with MLSHADE-SPA, DEGMCSO algorithm is better than it on 10 functions and slightly worse than it on 2 functions. Compared with MOS, the champion algorithm of CEC2012 competition version, DEGMCSO obtains better performance on 13 functions than MOS.

Table 4
Six Data sets from UCI.

Dataset	Features	Patterns	Class
movement	90	360	15
musk	167	6598	2
arrhythmia	279	452	16
madelon	500	2600	2
isolet5	617	1559	26
InterAd	1588	3279	2

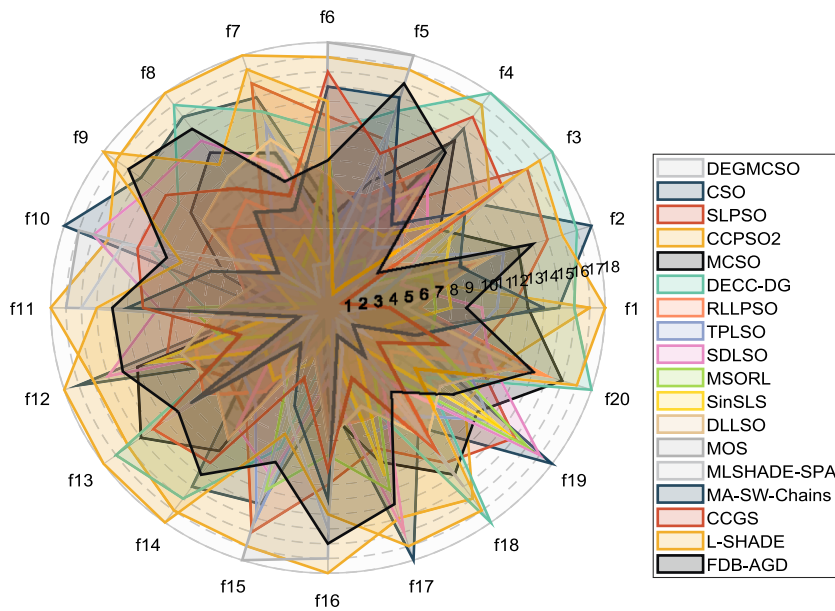


Fig. 5. Ranking of different algorithms in IEEE2010 test set.

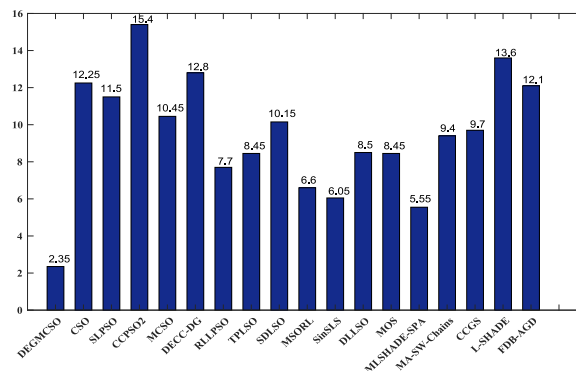


Fig. 6. Ranking of different algorithms in IEEE2010 test set.

On the basis of the average rank, DEGMCSO is at the top among 18 algorithms. DEGMCSO outperforms other comparison algorithm on CEC 2010 test set. The following in second and third place are MLSHADE-SPA and SinSLSO algorithm.

While DEGMCSO performs well for most functions in the CEC2010 test set, it performs worse than MOS and MLSHADE-SPA for other functions (e.g., f_{19} , f_{20}). For non-separable functions, the performance of DEGMCSO algorithm needs to be further improved, and better solutions can be found by introducing dimension-based decomposition or gene targeting techniques.

For functions f_1 – f_4 , f_8 – f_{11} , f_{12} and f_{15} , DEGMCSO can find a better solution in first half evolution and it is faster than other algorithms. The Fig. 5 shows that DEGMCSO has better search efficiency. For functions f_5 – f_7 , f_{13} and f_{14} , the algorithm DEGMCSO finds better solutions in the second half of the evolution. For convergence curve of Fig. 5, DEGMCSO always performs the best convergence performance on 15 functions, and can converge to a better solution faster.

4.4. The comparison of CEC'2013 test suite

To comprehensively evaluate the performance of DEGMCSO algorithm, in addition to SLPSO, CCPSO2, MCSO, DECC-DG, CSO, TPLSO, DLLSO, RLLPSO, MSORL, SDLPO and SinSLSO, we further compare DEGMCSO with the winner MOS of the CEC2013 competition, the two top algorithm MLSHADE-SPA and SHADE-ILS of the CEC2018 competition and excellent large-scale optimization algorithms CC-CMA-ES [40], L-SHADE and FDB-AGDE. To evaluate DEGMCSO, DEGMCSO and other 17 methods are adopted to handle functions of CEC2013 benchmark test suite. CEC2013 adds a class of overlapping functions with coupled modules. In addition, on the basis of the CEC 2010 benchmark, the CEC2013 introduces two new features, namely uneven module size and unbalanced module contribution. The experimental results were respectively compared by the convergence speed and numerical analysis. Table 3

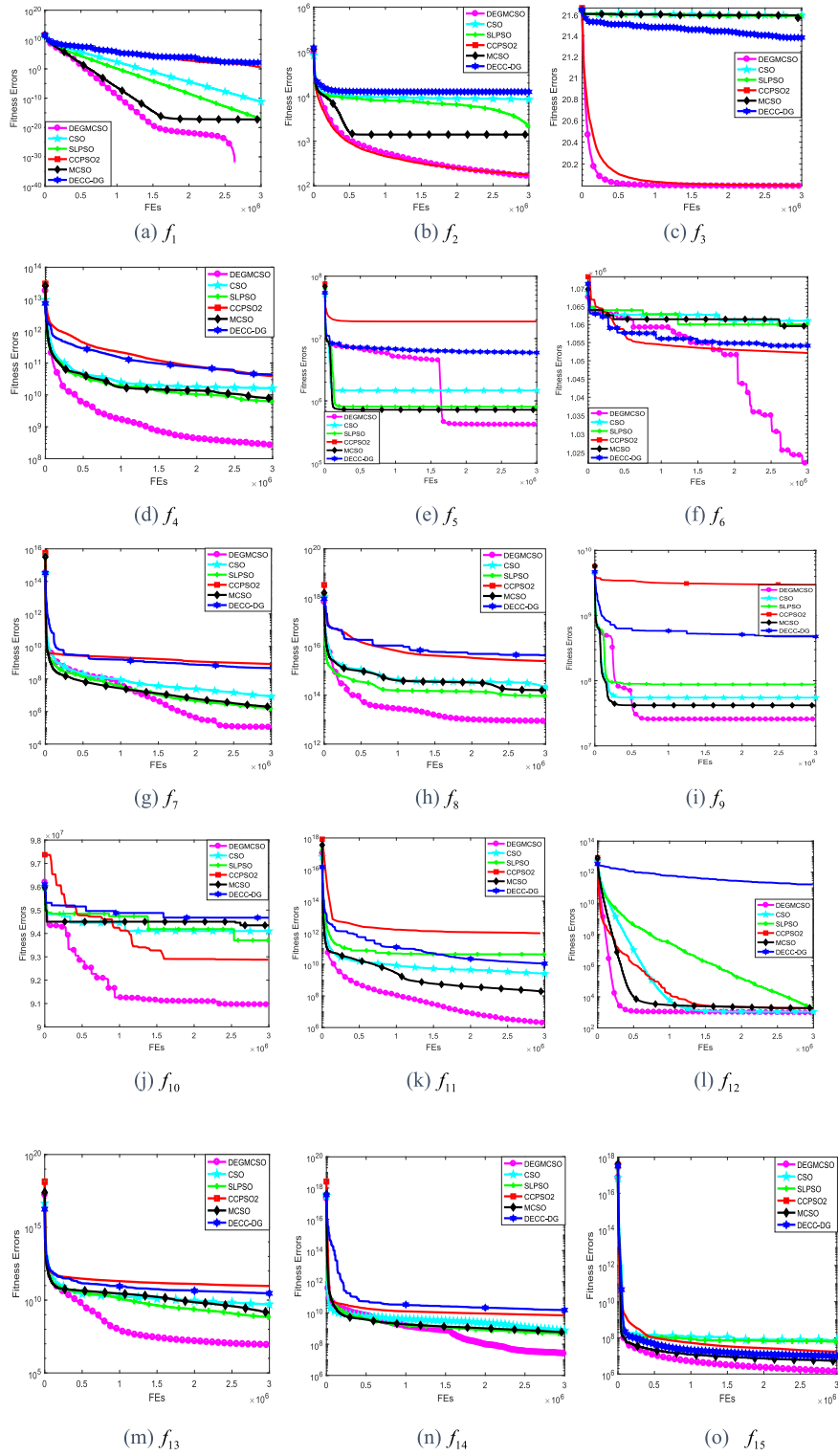


Fig. 7. Convergence curves in CEC2013 test suite.

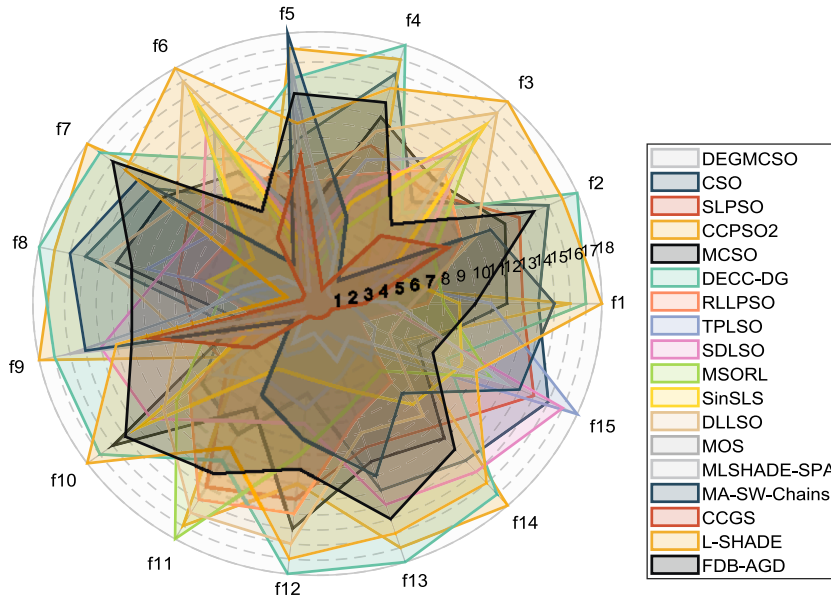


Fig. 8. Ranking of different algorithms in IEEE2013 test set.

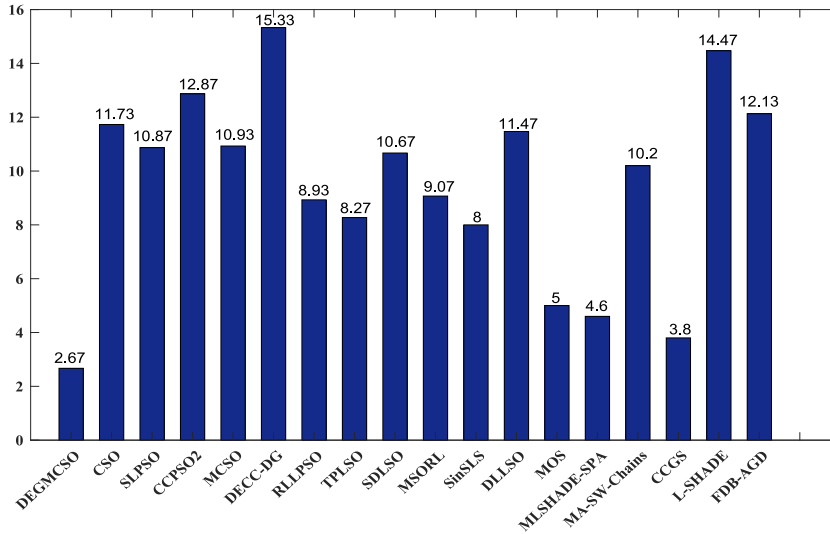


Fig. 9. Ranking of different algorithms in IEEE2013 test set.

shows the test results for $D = 1000$ dimensions. The values highlighted in bold are used to represent the best average results for each benchmark function. The experiments were conducted on CEC '2013 set. Table 3 records the results.

From Table 3, for CEC2013 test set, DEGMCSO is superior to SLPSO, CCPSO2, MCSO, DECC-DG, CSO, TPLSO, DLLSO, RLLPSO, MSORL, SDLPSO, SinSLS, MLSHADE-SPA, SHADE-ILS, CC-CMA-ES, L-SHADE and FDB-AGDE on 5 functions. For function f_2 , the improved algorithm is slightly worse than MLSHADE-SPA, but it is better than other algorithms. According to Friedman test, the rank of DEGMCSO is smaller than that of other 17 algorithms. In summary, the results on CEC2013 function test set show that the proposed algorithm DEGMCSO has higher solving accuracy than the other 17 optimization algorithms, and is superior to all comparison algorithms. According to rank comparison in the Friedman test, DEGMCSO and SHADE-ILS have good overall performance in handling a wide range of LSOPs. Compared with MOS, DEGMCSO still maintains good performance. DEGMCSO is better than MOS on 8 functions, and MOS is better than DEGMCSO on only 6 functions. DEGMCSO can achieve a similar or even outperforms the winner MOS. Compared with MLSHAD-SPA, DEGMCSO beats MLSHAD-SPA on 9 functions. DEGMCSO algorithm performs significantly better than MLSHAD-SPA on functions, especially on f_1 and f_5 . Compared to SHADE-ILS, DEGMCSO outperformed SHADE-ILS on 8 functions, but lost to SHADE-ILS in 6 functions. DEGMCSO has a similar performance to SHADE-ILS. The ranking result of each function in the function set is shown in the Fig. 8 and Fig. 9. The convergence curves of the comparison algorithm on different test functions are drawn

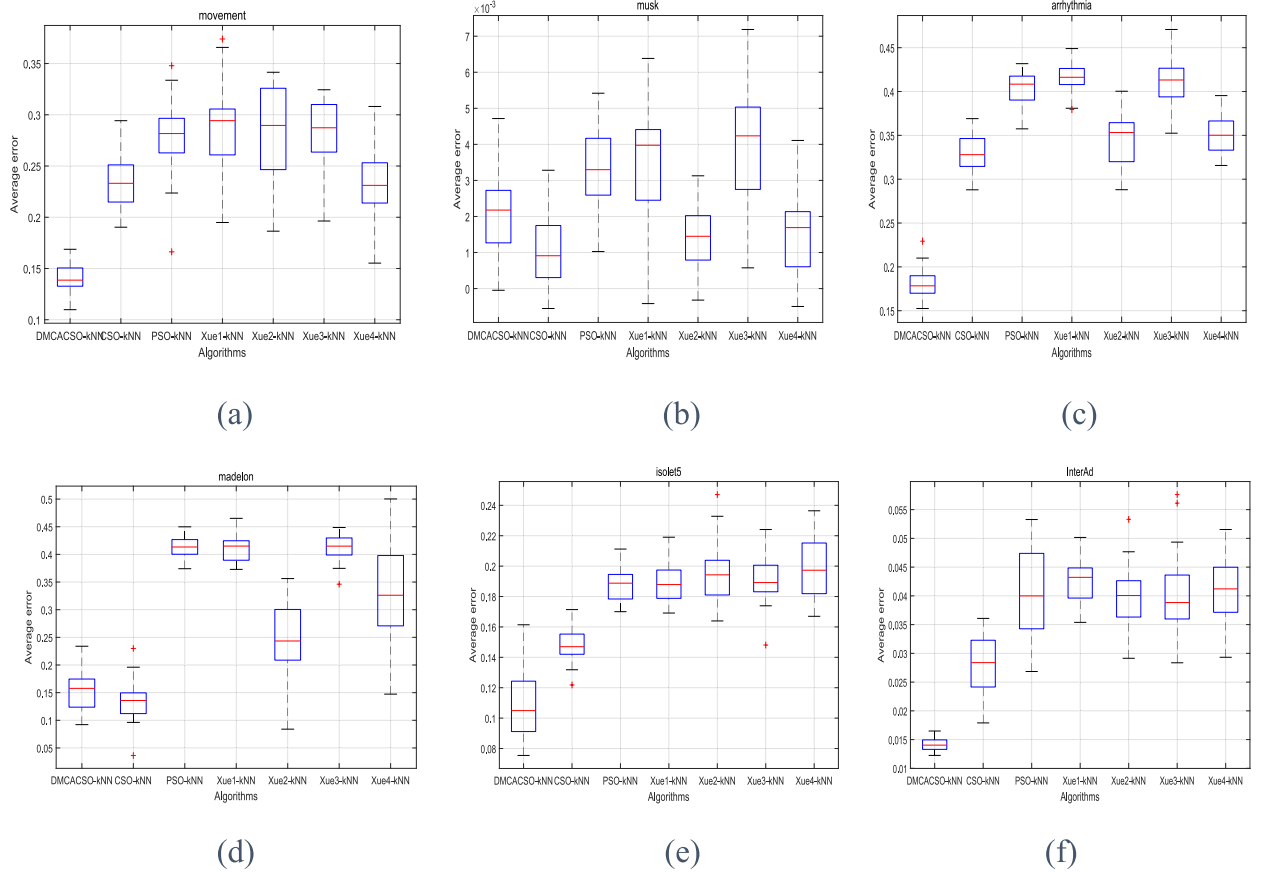


Fig. 10. Box-plot characteristics of seven algorithms.

in Fig. 7 and Fig 10.

5. DEGMCSO for the feature selection

5.1. Representation of solutions

The feature selection of data set based on DEGMCSO method is transferred into a binary optimization problem which limits the space to $\{0,1\}$. Therefore, the continuous optimization problem needs to be transformed into the discrete optimization problem when the feature selection is carried out.

The first step is to encode the particles in the population. A complete feature selection solution corresponds to the dimension of a particle. If $T(x_{ij}) = 1$, it indicates that the j -th feature of the i -th particle is selected; if $T(x_{ij}) = 0$, it indicates that the j -th feature of the i -th particle is not selected. The sigmoid function is used to convert continuous variables to the binary. The dimensions of the particles need to be mapped to $\{0, 1\}$, and x_{ij} need to be brought into the sigmoid function.

$$T(x_{ij}) = \frac{1}{1 + e^{-x_{ij}}} \quad (14)$$

Table 5
Average error.

Dataset	DMCACSO-kNN	CSO-kNN	PSO-kNN	Xue1-kNN	Xue2-kNN	Xue3-kNN	Xue4-kNN
movement	0.1376/0.0204	0.2394/0.0326	0.2850/0.0392	0.2850/0.0399	0.2896/0.0415	0.2832/0.0409	0.2369 /0.0327
musk	0.0023/0.0013	0.0011/0.0009	0.0034 /0.0016	0.0033/0.0017	0.0015/0.0009	0.0039/0.0019	0.0016/0.0009
arrhythmia	0.1814/0.0215	0.3240/0.0225	0.4059/0.0221	0.4076/0.0203	0.3593/0.0335	0.4098/0.0243	0.3564 /0.0294
madelon	0.1553/0.0412	0.1572/0.0370	0.4111/0.0189	0.4066/0.0210	0.2704/0.1006	0.4168/0.0230	0.3736/0.1027
isolet5	0.1127/0.0187	0.1491/0.0124	0.1875/0.0118	0.1855/0.0141	0.1917/0.0157	0.1922/0.0172	0.1954/0.0149
InterAd	0.0136/0.0015	0.0289/0.0044	0.0404/0.0071	0.0406/0.0058	0.0403/0.0055	0.0411/0.0056	0.0401/0.0050
Win/lose/tie		5/1/0	6/0/0	6/0/0	5/1/0	6/0/0	5/1/0

5.2. Application for feature selection

To study DEGMCSO's performance for solving practical application problems, we use six datasets from UCI library [41] in table 6 and k -nearest neighbor (kNN) classifier to confirm efficiency of DEGMCSO. k is set to 5. PSO and some variants from [42,43] are used to compare preference with DEGMCSO for feature selection. The statistical data are recorded in Table 5. The box diagram is shown in Fig. 8.

In terms of average values, the error rate based on DEGMCSO-kNN is smaller than all comparison methods on the movement, arrhythmia datasets, madelon, datasets, isolet5 datasets and InterAd datasets, respectively. For mean value and standard deviation, DEGMCSO-kNN algorithm performs significantly better than other methods. But for the musk dataset, the error rate of DMCACSO-kNN is higher than CSO-kNN, Xue2-kNN and Xue4-kNN.

6. Conclusion

Aim to improve the solving efficiency of CSO in the search process, a sample and efficient adaptive dual-criteria evaluation global best particle optimization strategy is presented in the paper. Through dual-criteria of multiple competitions, the particles that really need to update are chosen as the losers, which can better balance between diversity and convergence. Meanwhile, the quality improvement based on global best individual is developed to assist the search. DEGMCSO is verified on CEC2010 and CEC2013. The experimental results indicate that DEGMCSO method obtain optimal solution with higher precision. Finally, DEGMCSO is adopted to the feature selection. It is found that the improved method introduces distance information when calculating evaluation criteria, so DEGMCSO's computational expense is slightly more than that of CSO. The next step is to design a method with low computational complexity.

CRedit authorship contribution statement

Chen Huang: Writing – original draft, Methodology, Data curation. **Yingjie Song:** Writing – review & editing, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hongjiang Ma:** Writing – review & editing, Validation, Methodology, Formal analysis. **Xiangbing Zhou:** Visualization, Supervision, Resources, Formal analysis. **Wu Deng:** Writing – review & editing, Validation, Resources, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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